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constant K_P is related to the total pressure by the expression

a) $P = 24\ K_P$ b) $P = 8\ K_P$

c) $24\ P = K_P$ d) none of these

12. In which of the following equilibrium, K_P and K_C are not equal?

a) $2\ NO(g) \rightleftharpoons N_2(g) + O_2(g)$

b) $SO_2(g) + NO_2 \rightleftharpoons SO_3(g) + NO(g)$

c) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

d) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

13. If x is the fraction of PCl_5 dissociated at equilibrium in the reaction

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

then starting with 0.5 mole of PCl_5, the total number of moles of reactants and products at equilibrium is

a) 0.5 – x b) x + 0.5

c) 2x + 0.5 d) x + 1

14. The values of K_{P_1} and K_{P_2} for the reactions

$$X \rightleftharpoons Y + Z$$

$A \rightleftharpoons 2B$ are in the ratio 9 : 1 if degree of dissociation and initial concentration of X and A be equal then total pressure at equilibrium P_1, and P_2 are in the ratio

a) 36 : 1 b) 1 : 1

c) 3 : 1 d) 1 : 9

15. In the reaction,

$Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq) + 3OH^-(aq)$,

if the concentration of OH^- ions is decreased by ¼ times, then the equilibrium concentration of Fe^{3+} will

a) not changed

b) also decreased by ¼ times

c) increase by 4 times

d) increase by 64 times

16. Consider the reaction where $K_P = 0.5$ at a particular temperature

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

if the three gases are mixed in a container so that the partial pressure of each gas is initially 1 atm, then which one of the following is true

a) more PCl_3 will be produced

b) more Cl_2 will be produced

c) more PCl_5 will be produced

d) none of these

17. Equimolar concentrations of H_2 and I_2 are heated to equilibrium in a 1 litre flask. What percentage of initial concentration of H_2 has reacted at equilibrium if rate constant for both forward and reverse reactions are equal

a) 33% b) 66%

c) $(33)^2$ % d) 16.5 %

18. In a chemical equilibrium, the rate constant for the forward reaction is 2.5×10^2 and the equilibrium constant is 50. The rate constant for the reverse reaction is,

a) 11.5 b) 5

c) 2×10^2 d) 2×10^{-3}

19. Which of the following is not a general